The Argument the Papers Share

SO70 T235, October 12, 2026 (first written at T161). Replaces the consistency note G47.5.0 of September 22, which is kept in NOTES/AUDIT.

The chain of why, in nine steps

Each step answers the question the one before it raises.

1. **Why mathematics?** Everything that exists is ruled or random, and both are mathematical. Anything outside mathematics would be outside logic, and nothing could be reasoned about it. (A3; A0, A0a, A1, A2)
2. **Why simple laws?** Each bit of a complete description halves its probability, so the universe starts simple, laws and starting state together. Since every structure exists, the half is a count: one string in 2^k begins with any given k bits. (A2; A4)
3. **Why does complexity grow?** Minds like ours are complex, so complexity has to grow from that simple start. Many simple rules grow complexity: the Game of Life's one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any computation. What grows is organization; the description stays short. (A4 Section 4.7; A13; A2)
4. **How does it grow?** Growth takes steps, because causality takes steps: a cause produces its effect as the next state. (A13; A39; A41)
5. **Why steps and not a continuum?** A continuum has no next step, so nothing in it can cause anything. A length can be measured as a continuum, and its points do not cause it. (A39; A41; A39a)
6. **Why must change have a cause?** Change without a cause is outside logic and reason. Nothing could derive it, explain it or predict it, and allowing it undermines physics, science and reason itself. Random means a structure

with no shorter rule, not an event without a cause. (A39; A13; A4; A8)

7. **What is time, and which way does it run?** Time is the causal steps, so it runs forward, from simple to complex. Each step is a rule applied to rules, and the math rules themselves apply them, with no fire from outside (Hawking's question): there is no stuff; relations exist as mathematics, their nodes zero-width points, and what happens, and so what can be observed, is their transitions. Discreteness gives nothing a size, since a size implies a continuous surface; what it supplies is a next step. Space is how far the steps spread, and the speed of light is their base rate, shared by each body between moving and changing in place. Minds run the same way: a thought merges many states into one, so only the forward step is produced by the mind itself, and memories are built by steps already taken. (A13; A12)

8. **Why can't the whole history run backward?** An endless chain of steps has a first step and no last one, so there is no end to start from. (A13; A4; A0, A0a, A1; A7; A8; A12)

9. **Why don't we find ourselves in a reversed stretch?** A finite reversal must say which step it turns back from, and each bit of that address halves its weight; for a history with no shortcut, its short descriptions run the forward history first. A reversed brain would not produce its own states either, its next state being set by heat converging from outside. So we find ourselves in the forward history. (A13; A4; A0, A0a; A7; A8)

Each point needs the facts before it, not their reasons

The reason given for a point need not be true for the rest of the chain to hold, because each later point uses the fact, not the reason. Point 2 explains why the laws and the start are simple; points 3 to 8 need only that they are, and observation shows it: the laws we know fit on a page, and the early universe was

smooth to about one part in a hundred thousand. A reader who rejects the measure still faces points 3 to 8 (Gary, October 10, 17:05 and 17:11: "Even if that's not why we have simple laws, we still have simple laws and the rest of the argument holds up ... the reason doesn't need to be true for the rest of the chain to hold"). The direction of time does not hang on points 4 and 5 either: in continuous time a simple start would still have to grow complex minds, and an endless future would still have no last moment to run back from; what 4 and 5 add is that time comes in discrete steps. Point 9's cost uses the measure; point 8 does not. Each paper should state its points so that one contested link cannot take the rest down with it.

Where each paper stands

The hub: A4 argues all nine, with what exists, minds, one self and agency, each at about the depth the other papers give it. No paper cites another of these papers (Gary, October 12, 10:31); each makes its argument in place.
The essay: A39a tells all nine briefly for mathematicians (The Mathematical Intelligencer), from why everything is mathematics to why time runs forward.
What exists: A3 is the central paper; A0, A0a and A1 summarize it; A40 shows that the hard problem of consciousness assumes its conclusion; A30 and A21 apply the ontology to fiction and simulation.
The measure: A2 derives it, as one argument for a short-paper journal; A4 states it and applies it across physics; A38 counts the bits; A8 applies it to many worlds, A9 to a common generator, A16 to why minds prefer simple explanations.
Steps and the continuum: A39, A41 and the essay A39a.
Transitions: A41 says gravity, space and time can only be discrete causal transitions, rules applied to rules, and where a theory of everything must reside

(a rewriting system); A39 holds the mathematics of points, size and Zeno's runner; A13 (time), A12 (minds), A3 (what exists), A4 (the hub) and A0, A0a, A1, A2 and A39 state it in brief.

The search: A42 runs the first pass over the shortest rewriting rules. Some grow two-dimensional causal orders and none a four-dimensional one, and a rule that rewrites a few relations at a time can match our spacetime only at large scales.

Time and minds: A13 for time, A12 for consciousness in the transitions and forward processing, A7 for records.

Using parts of the chain: A5, A6, A15, A18, A28, A31.

Off the chain: A11 (alignment), A34 (economics), A35 (consensus), A36 (climate communication).

How the papers share the argument

Each paper states the points it relies on in a few sentences, with their reasons, so it stands on its own, and only those: a premise a paper does not use gives referees something to attack and supports nothing. Points 1 to 9 belong in A4 and A13, and in brief in the essay A39a; points 1 to 3 in A2; points 4 to 6 in A39 and A41; points 2, 6, 8 and 9 in A8 and A7; points 3, 4, 7 and 8 in A12; point 1 in A3 and the other ontology papers; none in A11, A34, A35 and A36. It cites none of the other papers, A4 included: each argument it uses is made in place, in a sentence or a paragraph. A philosophy paper that needs the hard problem of consciousness restates its one or two main paragraphs: the hard problem assumes that experience is not structure, and qualia theories, panpsychism, property dualism and idealism each owe an account of their extra ingredient that turns out to be more structure. Physics papers do not need it.